WIRSCHING'S WEAK COVERING CONJECTURE: AN EXTENDED COMPUTATION, A CONDITIONAL BOUND, AND THE STRUCTURE OF WHAT RESISTS IT

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Abstract. Wirsching's 1998 Weak Covering Conjecture asks for a sub-exponential $K(l)$ such that a large enough set $R_{j-1,j} \subset \mathbb{Z}_3^*$ is forced to cover every unit residue modulo 3^l. We extend the exact computation of $j^*(l)$, the smallest such budget, from $l = 20$ to $l = 23$, and compare four growth models for $e(l) := j^*(l) - l \log_4 3$ against the extension's tail. A constant model is unconditionally impossible (we prove $e(l) \geq \frac{1}{4}\log_2 l - O(1)$) and disfavored throughout; the growing alternatives are not distinguished by these criteria. Our best bound, $j^*(l) \leq \frac{9}{8}l + \frac{33}{2}$, comes from a mean-payoff-game relaxation resting on one correspondence, verified computationally but not proven in general. The natural Fourier route is blocked at its first step: its positivity criterion is provably unreachable at every budget; granting it regardless, square-root cancellation still could not beat the mean-payoff bound. Separately, null-model diagnostics at one tractable pair find structure neither construction forces, left uncalibrated. We prove that, for budgets $j \geq l$, the uncovered set at a budget is contained in both twice and four times the uncovered set at the previous budget, giving a recursive upper bound on $j^*(l)$. This bound is proven exact for $l+2 \leq j \leq j^*(l)$ at $l = 3, \dots, 13$ (given one further verified property), empirical beyond; the boundary $j = l+1$ is proven at $l = 3, \dots, 6$, empirical otherwise. We falsify a natural cost-1 repair rule it suggests, prove every last-uncovered residue is $\equiv 1 \pmod 3$ whenever $j^*(l) \geq l+1$ (all but one level), prove a two-class containment law mod 9 on these residues, and report further identities, some verified but not proven. Two open questions are named as exact conditions, each narrower than the conjecture; others arise where noted. The Weak Covering Conjecture remains open.

Keywords: Collatz map, 3-adic dynamics, covering conjecture, mean-payoff games, exponential sums, Littlewood–Offord theory.
2020 Mathematics Subject Classification: 11B50, 11K70, 91A50, 11L07, 37A44.

1. Introduction

Wirsching's 1998 monograph [1] studies the density of predecessor sets under the $3n+1$ map through an averaging construction on the 3-adic integers. A limiting Markov chain built from that construction has an invariant density, and Wirsching's own route to positive predecessor density passes through a covering question, stated in that same monograph [1], about a family of finite subsets of $\mathbb{Z}_3^*$, the 3-adic units. A later paper of his own isolates this route as a self-contained target and states, as open conjectures, the gaps still unbridged between it and a proof [2]. For integers $j, k \geq 0$, write

$$R_{j,k} := \left\{ \sum_{i=0}^{j} 2^{a_i} 3^i \;\middle|\; j+k \geq a_0 > a_1 > \cdots > a_j \geq 0 \right\} \subset \mathbb{Z}_3^*. \tag{1}$$

Each element of $R_{j,k}$ is built from a strictly decreasing sequence of $j+1$ "digit positions" chosen from $\{0, \dots, j+k\}$: the i-th chosen position, counting from the top, contributes 2^{a_i} scaled by 3^i. Wirsching's Weak Covering Conjecture (Conjecture 3.9, Chapter V, [1]) asks for a sub-exponential $K(l)$, meaning $\lim_{l\to\infty} K(l)e^{-\gamma l} = 0$ for every constant $\gamma > 0$ (Wirsching's own definition, same chapter), such that $|R_{j-1,j}| \geq K(l) \cdot 3^l$ already forces $R_{j-1,j}$ to cover every unit residue modulo 3^l; the cardinality on the left is that of the tuple family itself, before reduction, which by Proposition 2's bijection equals $\binom{2j}{j}$; the (a priori much smaller) size of its image modulo 3^l is a different quantity. Writing $j^*(l)$ for the smallest j such that $R_{j-1,j}$ covers $(\mathbb{Z}/3^l\mathbb{Z})^*$ exactly (a quantity that exists for every l, unconditionally; Proposition 16 below), the conjecture's forcing form and the growth rate of $j^*(l)$ coincide once coverage, once achieved, persists at every larger budget: this holds for $j \geq l$ (Theorem 10, Section 6), which covers every level in Table 1, but is not established below l, the regime the conjectured rate would eventually enter. With that proviso, the conjecture concerns the growth rate of

$$e(l) := j^*(l) - l\log_4 3, \tag{2}$$

the excess of the true covering budget over the leading-order rate a cardinality argument alone predicts ($|R_{j-1,j}| = \binom{2j}{j} \sim 4^j/\sqrt{\pi j}$, so matching $2\cdot 3^{l-1}$ residues needs $j \sim l\log_4 3$ at leading order). At the threshold, $K(l) \asymp |R_{j^*(l)-1,j^*(l)}|/3^l \asymp 4^{e(l)}/\sqrt{j^*(l)}$; the $\sqrt{j^*(l)}$ correction is polynomial in l at every level actually computed, where $j^*(l) = O(l)$ (Table 1), though no unconditional polynomial bound on $j^*(l)$ is proved here (Proposition 16's unconditional bound is exponential; Corollary 6's linear bound is conditional). This forward direction needs no persistence proviso: $j^*(l)$ is defined as a minimum, the conjecture (for any valid sub-exponential K) makes the first budget j with $\binom{2j}{j} \geq K(l)\cdot 3^l$ a covering budget, and a minimum is at most any member of its own defining set, so $j^*(l) \leq l\log_4 3 + o(l)$ follows outright; combined with the unconditional lower bound of Proposition 2, the conjecture's truth forces $e(l)$ to grow strictly slower than linearly, with no proviso attached. The proviso above belongs to the *equivalence* claimed at the start of this paragraph, not to this one-directional implication: constructing a valid K *from* the threshold density needs coverage to persist at every budget past $j^*(l)$, unproven below l. The converse, that $e(l) = o(l)$ would in turn establish the conjecture, needs the same persistence, which Theorem 10 only gives for $j \geq l$; below l, the regime a truly sub-linear $e(l)$ would eventually reach, the converse is open.

An earlier, unpublished manuscript of the author computed $j^*(l)$ exactly for $l = 1, \ldots, 20$ by direct enumeration and reported the resulting table without a growth-rate analysis. This paper does three things. First, it extends that table to $l = 23$ and gives a proper statistical comparison of candidate growth models for $e(l)$ against the extended table's tail (Sections 2 and 3). Second, it gives the best upper bound on $j^*(l)$ we have found, via a reformulation of the covering budget as a mean-payoff game, conditional on the empirically verified correspondence between the two stated in Empirical Result 4 (Section 4). Third, and at greatest length, it reports a sustained attempt to close the gap to the conjectured rate directly, first by the naive uniform-cancellation version of the natural Fourier approach, and, once that version is shown to fail for a checkable reason, by a direct study of the residues that actually resist coverage (Sections 5–7). That second attempt does not settle the conjecture. It produces a theorem on those residues, a falsification of the cost-1 form of a repair rule the theorem might suggest, and two named open questions, each narrower than the conjecture itself and worth a dedicated attack in its own right.

2. The computation

Table 1 gives $j^*(l)$ and $e(l)$ for $l = 1, \ldots, 23$. The values for $l \leq 20$ reproduce the earlier manuscript's table exactly; they were independently recomputed here by a from-scratch Rust reimplementation of the covering search (a bitset rotation dynamic program over residues modulo 3^l), validated against the original Python implementation at every $l \leq 20$ before being trusted for the extension. Each reported $j^*(l)$ is the smallest budget at which the DP's own coverage bitset is full, so the search certifies both coverage at $j^*(l)$ and its absence at $j^*(l)-1$ internally, by construction, at every level it runs; the $l \leq 20$ agreement against an independently written implementation is this paper's evidence that the search itself is correct. $l = 21$ has since been cross-checked the same way: a separate, from-scratch Python implementation (native bignum bitsets, following the Rust DP's recurrence as a specification rather than copying its code) independently reproduces $j^*(21) = 25$, confirming both that $j = 24$ fails to cover and that $j = 25$ does. $l = 22, 23$ are carried forward by inference only, without a separate cross-check at those levels; both are new.

The search above only tries budgets $j \geq l$ at level l, so on its own it certifies $j^*(l)$ only against smaller candidates within that range.

Lemma 1 (No smaller budget covers)**.** *For $l = 2, \ldots, 23$, no $j < l$ has $R_{j-1,j}$ covering $(\mathbb{Z}/3^l\mathbb{Z})^*$, so $j^*(l) \geq l$ holds outright, not only relative to the range the search above tries.*

Proof. If $R_{j-1,j}$ covers $(\mathbb{Z}/3^L\mathbb{Z})^*$ for some $L \geq j$, it covers $(\mathbb{Z}/3^j\mathbb{Z})^*$ too: reduction modulo 3^j factors through reduction modulo 3^L, and the ring homomorphism $\mathbb{Z}/3^L\mathbb{Z} \to \mathbb{Z}/3^j\mathbb{Z}$ restricts to a surjection of unit groups (every unit modulo 3^j lifts to a unit modulo 3^L), so the image of a full set of units modulo 3^L is a full set of units modulo 3^j.

$R_{0,1} = \{1, 2\}$ has 2 elements, short of $|(\mathbb{Z}/3^l\mathbb{Z})^*| = 2 \cdot 3^{l-1} \geq 6$ for $l \geq 2$, so $j = 1$ never covers at $l \geq 2$: $j^*(l) \geq 2$ for $l \geq 2$.

Fix l with $2 \leq l \leq 23$ and suppose, for contradiction, some j_0 with $2 \leq j_0 < l$ has R_{j_0-1,j_0} covering $(\mathbb{Z}/3^l\mathbb{Z})^*$. The reduction above, with $L = l$, gives R_{j_0-1,j_0} covering $(\mathbb{Z}/3^{j_0}\mathbb{Z})^*$ too, i.e. $j^*(j_0) \leq j_0$. But $j_0 < l \leq 23$, so j_0 is itself a level the search above already ran, starting from budget j_0 itself, and the search directly certifies that budget j_0 fails to cover $(\mathbb{Z}/3^{j_0}\mathbb{Z})^*$ there (this is the raw content of Table 1's $j^*(j_0) > j_0$ entry, independent of whether any smaller budget also fails, which is not assumed). Contradiction. So no $j_0 < l$ covers, giving $j^*(l) \geq l$. $\square$

l	$j^*(l)$	$e(l)$	l	$j^*(l)$	$e(l)$	l	$j^*(l)$	$e(l)$
1	1	0.208	9	13	5.868	17	21	7.528
2	4	2.415	10	15	7.075	18	22	7.735
3	6	3.623	11	16	7.283	19	23	7.943
4	7	3.830	12	17	7.490	20	24	8.150
5	9	5.038	13	18	7.698	21	25	8.358
6	10	5.245	14	19	7.905	22	26	8.565
7	11	5.453	15	20	8.113	23	27	8.773
8	12	5.660	16	20	7.320			

TABLE 1. $j^*(l)$ and $e(l) = j^*(l) - l \log_4 3$, $l = 1, \ldots, 23$.

The single plateau in the table, $j^*(15) = j^*(16) = 20$, is not an error: $l = 15 \to 16$ is the only step among the 22 from $l = 1$ to $l = 23$ at which the covering budget does not increase, and it is exactly the event Section 3's plateau count records. The integer difference $j^*(l) - l$, a coarser cousin of $e(l)$, is $0, 2, 3, 3, 4, 4, 4, 4, 4, 5, 5, 5, 5, 5, 5, 4, 4, 4, 4, 4, 4, 4, 4$ for $l = 1, \ldots, 23$: it never grows past 5 over the computed range, and nothing proven here rules out that ceiling persisting; Proposition 2's logarithmic bound on $e(l)$ does not touch it, since a bounded $j^*(l) - l$ makes $e(l) = (j^*(l) - l) + l(1 - \log_4 3)$ grow linearly, comfortably above a logarithmic floor. If the ceiling did persist, $j^*(l) = l + O(1)$ would make $e(l)$ linear, and, by the identity noted after (2), would make $K(l)$ exponential and falsify the conjecture, the contrapositive of the proviso-free forward direction stated there. This flatness is a fact about the range checked, not a proven bound either way.

$l = 21$ fits in physical RAM once a memory-wasteful parallelization step in the original implementation was corrected. $l = 22$ needs approximately 84 GiB of state, above the machine's 62 GiB of physical RAM, so a 500 GiB swap file was added to the machine specifically for this computation; the run took approximately 10,749 s (∼3 hours) with swap I/O as the dominant cost, against 748 s for $l = 21$ without swap. $l = 23$ required approximately 263 GiB of state, comfortably within the same swap budget, and 375,615 s (∼104 hours). Three attempts at $l = 24$, made over the course of this project on the same machine, each failed before completion (two to memory-policy kills, one to an apparent out-of-memory condition with no surviving kernel log confirming it directly); $l = 24$ is not attempted further, and the table stops at $l = 23$.

3. The growth of $e(l)$

We compare four candidate functional forms for $e(l)$'s growth against the tail $l = 10, \ldots, 23$ ($n = 14$; the same range choice as the earlier manuscript, now with three more points), each fit by ordinary least squares: $e(l) = c$ (constant), $e(l) = a + b \ln l$ (logarithmic), $e(l) = a + b\sqrt{l}$ (square-root), and $e(l) = a + bl$ (slow-linear). Table 2 gives the fit statistics, using the Gaussian-OLS forms $\mathrm{AIC} = n \ln(\mathrm{RSS}/n) + 2k$, $\mathrm{BIC} = n \ln(\mathrm{RSS}/n) + k \ln n$ (RSS the residual sum of squares, k the number of regression coefficients, 1 for the constant model and 2 for the other three), and leave-one-out cross-validated RMSE; since only differences ΔAIC, ΔBIC are used below, the convention for a shared additive constant across models does not affect them. $e(l)$ is an exact, deterministic sequence with no sampling model behind it, so these are read descriptively throughout, as penalized fit scores over the observed table rather than calibrated statistical evidence.

model	k	ΔAIC	ΔBIC	LOOCV RMSE
constant	1	16.091	15.452	0.5200
logarithmic	2	1.512	1.512	0.2991
square-root	2	0.722	0.722	0.2909
slow-linear	2	0.000	0.000	0.2837

Table 2. Model comparison on $e(l)$, $l = 10, \ldots, 23$. ΔAIC/BIC relative to the best-fitting model (slow-linear).

A constant model is disfavored sharply and consistently: $\Delta\mathrm{AIC} = 16.09$, $\Delta\mathrm{BIC} = 15.45$, both growing at every l added since $l = 21$. The three growth models are not distinguished from one another by the conventional $\Delta\mathrm{AIC} > 2$ rule of thumb, with slow-linear taking a narrow lead on both AIC and leave-one-out cross-validation; the three candidate regressors ($\log l$, $\sqrt{l}$, l) are themselves highly correlated over this range (pairwise correlation ≥ 0.993), so this ranking mainly reflects how thin the current margin is. By the identity noted after (2), a linear $e(l)$ would make $K(l)$ exponential, falsifying the conjecture outright. Fourteen highly correlated points cannot distinguish that possibility from logarithmic or square-root growth, so this comparison does not settle the question either way; it only says which growth rates remain live.

None of this machinery is actually needed to rule out a constant $e(l)$: since $\log_4 3 < 1$, rounding a constant-plus-linear quantity $l \log_4 3 + c$ to the nearest integer can only change by 0 or 1 from one l to the next, yet $j^*(1) = 1$ and $j^*(2) = 4$, an increment of 3. A constant-$e(l)$ source is incompatible with the table's first two entries alone, with no statistical test required.

A direct combinatorial fact rules out the same tail even more sharply, restricted to where the model is not already dead over the full table: the constant-rounding model behind it predicts every increment in $\{0, 1\}$, which the full table already refutes (the increment of 3 noted above, and increments of 2 elsewhere in the table, e.g. $j^*(2) = 4$ to $j^*(3) = 6$). Restricted to the $l = 10, \ldots, 23$ tail the model comparison above uses (13 increments over the same 14 points, all actually in $\{0, 1\}$, consistent with the model's own prediction there), a fixed-c model $J_l := \mathrm{round}(l \log_4 3 + c)$ forces the total change over any 13 consecutive steps into $\{\lfloor 13 \log_4 3 \rfloor, \lceil 13 \log_4 3 \rceil\} = \{10, 11\}$, for every c (a standard fact about round of an arithmetic progression with irrational slope, checked directly over a fine grid of c). The observed change is $j^*(23) - j^*(10) = 27 - 15 = 12$, outside that pair for every c: no constant-rounding model can reproduce this tail at all, a conclusion that needs no probability. $e(l)$ is an exact, deterministic sequence throughout, so nothing in this paragraph carries a sampling interpretation. Among the 13 increments, exactly one is zero (the $l = 15 \to 16$ plateau noted above); under the constant-rounding model, $e(l+1) - e(l)$ can only be $1 - \log_4 3 \approx 0.2075$ or

$-\log_4 3 \approx -0.7925$. The plateau count above and the OLS comparison above are both functions of the same 14-point increment sequence, so they are not independent checks, though the plateau observation bears specifically on the constant-rounding model, a different object from the free-intercept constant regression $e(l) = c$ that Table 2 fits.

A constant $e(l)$ is disfavored descriptively by every measure above. It is also, unconditionally, impossible.

Proposition 2 (Unboundedness of $e(l)$)**.** $e(l) \geq \frac{1}{4}\log_2 l - O(1)$.

Proof. Distinct exponent tuples give distinct sums: since the a_i are distinct, a_{j-1} is the unique minimal one, so $V = \sum_i 2^{a_i}3^i$ has 2-adic valuation exactly a_{j-1} (every other term is divisible by a strictly higher power of 2), recovering a_{j-1} from V directly. The full term $2^{a_{j-1}}3^{j-1}$ is then subtracted off, and the process iterates on the remaining $j-1$ terms. So $R_{j-1,j}$, in bijection with j-subsets of the $2j$-element set $\{0,\ldots,2j-1\}$, has exact cardinality $\binom{2j}{j}$ (this cardinality is Wirsching's own Corollary 1.8, Chapter V, [1]; the injectivity argument above is elementary and appears already, for the same 2^a3^b-type sums, in [8]). Covering $(\mathbb{Z}/3^l\mathbb{Z})^*$, of size $2\cdot 3^{l-1}$, requires the domain to be at least as large as the codomain: $\binom{2j}{j} \geq 2\cdot 3^{l-1}$ is necessary at $j = j^*(l)$ in particular, since $R_{j^*(l)-1,j^*(l)}$ covers by definition of $j^*(l)$. The explicit bound $\binom{2n}{n} \leq 4^n/\sqrt{3n+1}$, valid for every $n \geq 1$, turns this into $4^j \geq 2\cdot 3^{l-1}\sqrt{3j+1}$, hence

$$j \geq l\log_4 3 - \log_4 3 + \log_4 2 + \tfrac{1}{2}\log_4(3j+1).$$

Dropping the last, non-negative term already gives $j \geq l\log_4 3 - O(1)$, so $3j+1 = \Omega(l)$, hence $\log_4(3j+1) \geq \log_4 l - O(1) = \frac{1}{2}\log_2 l - O(1)$. Feeding this back into the displayed inequality gives the claimed bound. $\square$

4. A mean-payoff-game upper bound

The full-precision game. Fix $l \geq 1$. A *full-precision play* from a unit z_0 modulo 3^l is a sequence of l moves: at step $i = 0,\ldots,l-1$, from a unit z_i modulo 3^{l-i}, a move of cost $d_i \geq 0$ is *legal* when $2^{d_i}z_i \equiv 1 \pmod 3$, and produces $z_{i+1} := T_{d_i}(z_i) := (2^{d_i+1}z_i - 2)/3$, which is then required to be a unit modulo 3^{l-i-1} (automatic at $i = l-1$, where $3^{l-i-1} = 1$). Write $J := \sum_{i=0}^{l-1} d_i$ for the total cost.

Lemma 3 (Grounding)**.** *For a legal full-precision play from z_0 with costs $d_0,\ldots,d_{l-1}$ and total cost J, define $a_{l-1} := 0$ and $a_{i-1} := a_i + d_i + 1$ for $i = l-1,\ldots,1$ (so $a_0 = J - d_0 + l - 1$). Then*

$$2^{J+l-1}z_0 \equiv \sum_{i=0}^{l-1} 2^{a_i}3^i \pmod{3^l}.$$

The right side is, by construction, an element of $U(l,a_0)$ (Section 7) with bottom exponent $a_{l-1} = 0$.

Proof. By induction on the play length n ($n = 1,\ldots,l$), stated for a play of length n starting at a unit z modulo 3^n (matching the lemma's notation at $n = l$, $z = z_0$). Write $J_n := \sum_{i=0}^{n-1} d_i$ and $a_{n-1} := 0$, $a_{i-1} := a_i + d_i + 1$ for $i = n-1,\ldots,1$, as above.

Base case $n = 1$: the single move has $2^{d_0}z \equiv 1 \pmod 3$, which is exactly $2^{a_0+d_0}z \equiv 2^{a_0}3^0 \pmod 3$ since $a_0 = 0$ when $n = 1$.

Inductive step: suppose the claim holds at length $n-1$ for the sub-play with costs $d_1,\ldots,d_{n-1}$ starting at $z' := T_{d_0}(z)$, a unit modulo 3^{n-1}; writing $a'_i := a_{i+1}$ for this sub-play, $a'_{n-2} = a_{n-1} = 0$ matches its base case, and its recursion $a'_{i-1} = a'_i + d_{i+1} + 1$ matches $a_i = a_{i+1} + d_{i+1} + 1$ (the original recursion at index $i+1$), so the induction hypothesis reads

$$2^{a_1+d_1}z' \equiv \sum_{i=0}^{n-2} 2^{a_{i+1}}3^i \pmod{3^{n-1}}.$$

By definition of T_{d_0}, $2^{d_0+1}z \equiv 3z' + 2 \pmod{3^n}$. Multiplying by the unit $2^{a_1+d_1}$ gives

$$2^{a_1+d_1+d_0+1}z \equiv 3 \cdot 2^{a_1+d_1}z' + 2^{a_1+d_1+1} \pmod{3^n}.$$

Since $a_0 = a_1 + d_1 + 1$ (the recursion at $i = 1$), the left side is $2^{a_0+d_0}z$ and the last term is 2^{a_0}. Multiplying the induction hypothesis by 3 turns it, modulo 3^n, into $3 \cdot 2^{a_1+d_1}z' \equiv \sum_{i=1}^{n-1} 2^{a_i}3^i$, so

$$2^{a_0+d_0}z \equiv 2^{a_0} + \sum_{i=1}^{n-1} 2^{a_i}3^i = \sum_{i=0}^{n-1} 2^{a_i}3^i \pmod{3^n},$$

the claim at length n, using $a_0 + d_0 = J_n + n - 1$ (telescoping the recursion). □

Empirical Result 4 (The game's worst case is $j^*(l)$).

$$j^*(l) = \max_{z_0} \min\{J : z_0 \text{ has a legal full-precision play of total cost } J\},$$

the maximum over units z_0 modulo 3^l. Verified against the known exact table of $j^*(l)$ for $l = 1, \ldots, 12$, with no discrepancy; that a legal full-precision play exists at all from every unit z_0 (the minimum inside the display is over a nonempty set) is part of what this verification confirms empirically. A general proof would additionally need the successor to stay a unit at the next modulus at each step, a condition the mod-3 legality check alone does not decide. Lemma 3 does not by itself establish this: the twist 2^{J+l-1} in the lemma depends on z_0 through its own optimal J, so it is not a single, z_0-independent relabeling of the unit group, and we do not have a general proof of the displayed identity.

The window-k relaxation. A move of cost $d \geq 0$ at z is *legal* when $2^d z \equiv 1 \pmod 3$, as above, and *safe* at window k when, for $m := 3^k$, every one of the three residues $z, z+m, z+2m$ modulo $3m$ maps under T_d back into a unit modulo m. A minimizing player, seeing only $z \bmod 3^k$, chooses a legal, safe d; an adversary then reveals the hidden next 3-adic digit $\epsilon \in \{0, 1, 2\}$, forcing the true state to $T_d(z + 3^k\epsilon) \bmod 3^k$. Both requirements matter: legality alone can send a unit to 0 (take $z = 1$, $d = 0$: $T_0(1) = 0$), and the safety filter is exactly what keeps the walk inside $\mathbb{Z}_3^*$ regardless of which digit the adversary supplies. Legality depends on d only through $d \bmod 2$ (since 2 has order 2 modulo 3), and 2 is a primitive root modulo 3^{k+1} for every $k \geq 0$ (order $2 \cdot 3^k$, by an elementary lifting-the-exponent induction), so $2^{d+1} \bmod 3^{k+1}$, and hence $T_d(z) \bmod 3^k$ and the safety check itself, depend on d only through $d \bmod (2 \cdot 3^k)$: successor states repeat with this period as d grows, and every representative beyond the smallest in its residue class has strictly larger cost for the same effect, so is never used by an optimal policy. The action set at each state is therefore, without loss, the subset of $\{0, \ldots, 2 \cdot 3^k - 1\}$ that is legal and safe there, which is finite. That this subset is nonempty at every unit state has been verified computationally at every $k = 3, \ldots, 14$ used below; a general proof is left open here.

This restricted-visibility game is, by construction, a two-player *mean-payoff game* on the $2 \cdot 3^{k-1}$ unit states modulo 3^k, with the now finite legal, safe moves as the minimizer's edges, the adversary's three choices of ϵ as the maximizer's edges, and payoff the long-run average of the move costs d. The classical theory of such games, initiated by Ehrenfeucht and Mycielski [5], guarantees a value and an optimal positional policy in general; Theorem 5 below needs none of that generality, only the twelve specific policies whose resulting (ρ_k, C_k) are reported in Table 3, each independently checked as described in Section 9.

A window-k policy, run against the actual digit sequence, gives a legal full-precision play at any z modulo 3^l: at step i, the true remaining state is z_i modulo 3^{l-i}, so the policy sees the true $z_i \bmod 3^k$ directly whenever $l - i \geq k$, i.e. for every step but the last $k - 1$. Once $l - i < k$, whether because $l < k$ globally or because a play with $l \geq k$ has reached its own final $k - 1$ steps, fix one lift of the true state to a residue modulo 3^k, at the first such step, by padding the true state's remaining $l - i$ digits with $k - (l - i)$ arbitrary further digits, and run the window-k game forward from that lift on the play's own remaining digits (not a fresh lift at each step). Two separate facts keep this consistent with the true state. First, safety, by construction, keeps every one of the three possible one-digit extensions of a safe state a unit modulo 3^k, so the window-k policy always has a well-defined next state $s_{i+1} := T_d(s_i + 3^k\epsilon_i) \bmod 3^k$ to act on for any extending digit $\epsilon_i \in \{0, 1, 2\}$, real or padding, matching the general successor $s'(\epsilon)$ used in the theorem's proof below. Second, from the defining formula $T_d(z) = (2^{d+1}z - 2)/3$, $T_d(z) \bmod 3^{k'}$ depends only on $z \bmod 3^{k'+1}$ for every $k' \geq 0$; since $k > l - i - 1$ throughout this regime, the term $3^k\epsilon_i$ never affects any digit below position k, in particular none of the $l - i$ low-order digits of the input s_i, which by the precision fact are exactly what determine the $l - i - 1$ low-order output digits this step tracks, so which ϵ_i is used, real when the true state still supplies one or arbitrary padding once it does not, is irrelevant to them. Consequently, if the window-k state s_i agrees with the true z_i on its lowest $l - i$ digits, as it does by construction at the first padded step and by induction thereafter, then s_{i+1} agrees with the true z_{i+1} on its lowest $l - i - 1$ digits regardless of ϵ_i, one fewer digit of guaranteed agreement each step, exactly matching how many true digits remain. Legality of the chosen move at step i depends only on $z_i \bmod 3$, the lowest digit, which is among the true, agreeing digits at every one of these steps since $l - i \geq 1$ throughout (steps run $i = 0, \ldots, l - 1$); at the terminal state z_l, reached after the last step, $l - i = 0$ and the unit requirement modulo $3^0 = 1$ is vacuous. So the chosen move stays legal against the true, lower-precision state at every step and the successor relation used below holds at every step, including these last $k - 1$, with ϵ_i taken to be the real next digit whenever the true state still has one and an arbitrary fixed digit thereafter. Writing s_i for whichever window-k state σ_k sees this way at step i (the true $z_i \bmod 3^k$, or a fixed lift of it), s_{i+1} is by construction the successor $s'_i(\epsilon_i)$ under σ_k's chosen move, so $s_0, \ldots, s_l$ is exactly a play of the form the proof below sums over, with move costs equal to the real play's (this regime is part of what the policy-against-the-covering-DP check at $k = 4, 7$ against l up to 14 already exercised, with zero illegal or unsafe moves; every play there passes through this same final-$k-1$-steps regime regardless of how large l is).

Theorem 5 (Window relaxation bound, conditional on Empirical Result 4)**.** *For each $k = 3, \ldots, 14$, the certified policy underlying Table 3 (a legal, safe move at every unit state modulo 3^k, average cost ρ_k) gives an explicit rational C_k with $j^*(l) \leq \rho_k \cdot l + C_k$ for all l.*

Proof. Fix k in range. The policy σ_k, the value ρ_k, and a potential $h_k : (\mathbb{Z}/3^k\mathbb{Z})^* \to \mathbb{Q}$ are computed by a nested Howard strategy-improvement solver searching over the finite action set established above, and σ_k's chosen move is checked, state by state and digit by digit, against

$$d_{\sigma_k}(s) + h_k(s'(\epsilon)) \leq \rho_k + h_k(s)$$

at every state s and every adversary digit ϵ, where $s'(\epsilon)$ is the successor under σ_k's chosen move. Summing the inequality along any l-step play $s_0, \ldots, s_l$ telescopes to

$$\sum_{i=0}^{l-1} d_{\sigma_k}(s_i) \leq l\rho_k + h_k(s_0) - h_k(s_l) \leq l\rho_k + \big(\max h_k - \min h_k\big) =: l\rho_k + C_k,$$

regardless of which digits the adversary actually supplies, since the potential inequality holds for every ϵ at every step. So σ_k, run as above, gives every unit z modulo 3^l a legal full-precision play of total cost at most $l\rho_k + C_k$; the minimum cost over all legal plays from z is therefore also at most $l\rho_k + C_k$, hence so is the worst case over z. Only the $j^*(l) \leq \max_{z_0} \min\{J\}$ direction of Empirical Result 4's equality is used here, to identify that worst case with $j^*(l)$; the reverse direction plays no role in this proof. □

σ_k, h_k and ρ_k are computed as rationals by a nested Howard strategy-improvement solver, which searches actions d up to a fixed cap of 40 (well above the largest d any certified policy below actually uses at any $k = 3, \ldots, 14$, which is 11). The potential inequality above is checked only for σ_k's chosen move at each state, against all three adversary digits, never against the rest of the action set; that chosen move's own legality and safety are verified directly from the full-game definitions above ($2^d z \equiv 1 \pmod 3$ and the three-lift condition), which do not reference the cap at all, so the resulting upper bound $j^*(l) \leq \rho_k l + C_k$ does not depend on it either. The solver also extracts a matching adversary lower bound from the same run, but that computation ranges only over the same capped action set: "upper bound equals lower bound" certifies ρ_k exactly as the value of the mean-payoff game restricted to $d \leq 40$, not, without further argument, as the value of the unrestricted game on the full action set $\{0, \dots, 2 \cdot 3^k - 1\}$ described above, since a move with $d > 40$, unavailable to the capped lower-bound computation, could in principle let the minimizing player achieve a lower true value than the capped certificate rules out. Table 3 reports ρ_k, C_k under this understanding: certified rates of an exhibited, verified policy, which is all Theorem 5 needs, not a proven-exact value of the full window-k game. σ_k and ρ_k are cross-validated by an independent value-iteration solver and, at $k = 4$ and $k = 7$, by running the resulting policy against the actual finite covering DP for every unit residue modulo 3^l, l up to 14, with zero illegal or unsafe moves encountered. Table 3 gives the values computed for $k = 3, \dots, 14$.

k	3	4	5	6	7	8	9	10	11	12	13	14
ρ_k	2	$\frac{5}{3}$	$\frac{3}{2}$	$\frac{3}{2}$	$\frac{7}{5}$	$\frac{25}{19}$	$\frac{5}{4}$	$\frac{11}{9}$	$\frac{6}{5}$	$\frac{7}{6}$	$\frac{119}{104}$	$\frac{9}{8}$
C_k	5	$\frac{19}{3}$	$\frac{15}{2}$	9	10	$\frac{207}{19}$	$\frac{47}{4}$	$\frac{115}{9}$	$\frac{69}{5}$	$\frac{44}{3}$	$\frac{1621}{104}$	$\frac{33}{2}$

TABLE 3. The certified window-relaxation rate ρ_k and constant C_k, $k = 3, \dots, 14$.

At $k = 14$, $\rho_{14} = 9/8 = 1.125$ and $C_{14} = 33/2$, giving:

Corollary 6 (Conditional on Empirical Result 4)**.** $j^*(l) \leq \frac{9}{8} l + \frac{33}{2}$.

This is, to our knowledge, the best bound on $j^*(l)$'s growth rate currently available, conditional on Empirical Result 4. ρ_k is observed to be non-increasing over the computed range $k = 3, \dots, 14$ (Table 3); we have not verified this in general, and Corollary 6 uses only the single value $k = 14$. The general technique, representing a covering-type combinatorial problem as a mean-payoff game and computing exact rational relaxation bounds this way, is not new to this application: beyond Ehrenfeucht and Mycielski's foundational theorem [5] itself, a close and recent methodological precedent proves rationality and computability of the covering radius of a primitive sofic shift by a mean-payoff-game reduction of the same kind, applied to a different object [6]; that reduction uses non-alternating mean-payoff games, introduced by the same authors in [7], a variant of the classical alternating games of [5] used here. We cite all three as precedent for the framing; what is new here is its specific application to $j^*(l)$.

Remark 7. Fitting $\rho_k = L + A/k$ by ordinary least squares to $k = 10, \dots, 14$ gives $L \approx 0.876$, still well above the conjectured true rate $\log_4 3 = 0.7925\dots$; the fit is sensitive to the range used (all of $k = 3, \dots, 14$ gives $L \approx 0.904$) and is not evidence of any specific limit. Only the direction $j^*(l) \leq \rho_k l + C_k$, proven above, is established. The reverse inequality, $\inf_k \rho_k \leq \limsup_l j^*(l)/l$, which would be needed to conclude anything about where ρ_k is heading, is not established.

5. THE EXPONENTIAL SUM, AND WHY IT RESISTS A DIRECT ATTACK

Corollary 6's conditional bound, $1.125\, l$, is well above the conjectured rate $\log_4(3)\, l \approx 0.7925\, l$. The natural route to closing that gap is Fourier-analytic. Here $e(x) := e^{2\pi i x}$ denotes the standard additive character, unrelated to the sequence $e(l)$ of (2). By orthogonality on $\mathbb{Z}/3^l\mathbb{Z}$, the number of tuples landing in residue class z is

$$N(z) = \frac{1}{3^l} \sum_{t=0}^{3^l - 1} S(t)\, e\left(-\frac{tz}{3^l}\right), \qquad S(t) := \sum_{j+k \geq a_0 > \dots > a_j \geq 0} e\left(\frac{t}{3^l} \sum_{i=0}^{j} 2^{a_i} 3^i\right), \tag{3}$$

and $S(0) = |R_{j,k}| =: T$, since $S(0)$ counts exponent tuples and the injectivity argument in the proof of Proposition 2 above (distinct tuples give distinct sums, recovering a_j from the sum's 2-adic valuation) applies unchanged to general (j,k), not only $(j-1,j)$. The natural route to positivity is the triangle inequality: $N(z) > 0$ for every z as soon as $\sum_{t\neq 0} |S(t)| < T$. This route is closed before it opens. Proposition 8 below shows the sum on the left is never below T, for any l, j, k: the criterion cannot be reached at all, for any budget. The remainder of this section studies what is left once the full-spectrum route is abandoned: a primitive-frequency version of the same idea (§5.2), and the further reasons below why it resists too.

5.1. **A hard ceiling on uniform cancellation.**

Proposition 8 (The full-spectrum criterion is unreachable)**.** *For every $l \geq 1$ and every $R_{j,k}$ with $T := |R_{j,k}| > 0$, $\sum_{t\neq 0} |S(t)| \geq T$; in particular $|S(3^{l-1})| \geq T/2$. Consequently the criterion $\sum_{t\neq 0} |S(t)| < T$ is never satisfied, for any l, j, k, and neither is the uniform bound $|S(t)| = o(T/3^l)$ at every nonzero t, since that would force $\sum_{t\neq 0} |S(t)| = o(T)$.*

Proof. Every element of $R_{j,k}$ reduces to 2^{a_0} mod 3 (all other terms carry a positive power of 3), so it is $\equiv 1$ or $2 \pmod 3$: no element is divisible by 3. Hence $N(z) = 0$ for every z divisible by 3, in particular $N(0) = 0$, so $\sum_{t=0}^{3^l-1} S(t) = 3^l N(0) = 0$ and $\sum_{t\neq 0} S(t) = -S(0) = -T$; the triangle inequality gives $\sum_{t\neq 0} |S(t)| \geq \left|\sum_{t\neq 0} S(t)\right| = T$.

The same collapse localizes: summing the inversion formula over the 3^{l-1} residues divisible by 3 leaves only the three frequencies $t \in \{0, 3^{l-1}, 2\cdot 3^{l-1}\}$ in the inner sum (every other character is nonconstant on that coset of the index-3 subgroup and averages to 0), giving $S(3^{l-1}) + S(2\cdot 3^{l-1}) = -T$ exactly, at every l, j, k. Since $S(3^l - t) = \overline{S(t)}$, the two terms are conjugate, so $\operatorname{Re} S(3^{l-1}) = -T/2$ exactly and $|S(3^{l-1})| \geq T/2$: a single conjugate pair already forces the bound, before the rest of the spectrum is added in. ☐

Proposition 8 closes the naive route outright, at every j: there is no threshold past which it starts to work. Even granting the unreachable premise for comparison, specializing $S(t)$, T to the family $R_{j-1,j}$ this paper's covering question concerns ($T = \binom{2j}{j}$) and asking what j a uniform bound $|S(t)| \leq C\sqrt{T}$ (square-root cancellation), applied via the triangle inequality across all $3^l - 1$ nonzero frequencies, would force gives $(3^l-1)C\sqrt{T} < T$, so $T > ((3^l-1)C)^2$, i.e. $4^j/\sqrt{3j+1} \geq T > ((3^l - 1)C)^2$ (the explicit bound from Proposition 2's proof), hence $j \geq 2\log_4(3)\, l - O(1) \approx 1.585\, l - O(1)$: already *worse* than Corollary 6's $1.125\, l$, on top of resting on a premise Proposition 8 rules out at every j. Both facts together rule out the naive version of the Fourier approach, and direct attention to whether a finer, frequency-dependent treatment (a major/minor-arc split, in the language of the circle method) can do better.

5.2. **A computation: the exceptional mass does not stay small.** Numerically, $|S(t)|$ is far from uniform. Writing $t = 3^{l-c}t'$ with $3 \nmid t'$ (valuation $l - c$), $t \cdot V/3^l = t' \cdot V/3^c$ for every tuple value V, so $S(t)$ at modulus 3^l equals the same sum evaluated at the coarser modulus 3^c: a frequency of valuation $l - c$ is literally the level-c sum of the same tuple family. Proposition 8's exact bound $|S(3^{l-1})| \geq T/2$ is only a lower bound, no claim of maximality: at $m = 1$, $l = 2$ (the family $R_{0,1} = \{1, 2\}$, $T = 2$), $|S(3)| = 1$ but $|S(1)| = 2\cos(\pi/9) \approx 1.879$ exceeds it, so the top pair is not always the largest frequency. At the levels directly checked in the family this section actually uses ($R_{m-1,m}$ at each level's own first-order cardinality threshold: $l = 10, m = 9$; $l = 12, m = 11$; $l = 14, m = 13$), the top pair $t = 3^{l-1}, 2\cdot 3^{l-1}$ is observed to be the largest, and high valuation correlates with large magnitude beyond it, but loosely: among the twelve largest frequencies at $l = 10$, valuations run $9, 9, 8, 8, 7, 7, 8, 8, 8, 8, 7, 7$, and at $l = 12$ the largest primitive (valuation-0) frequency already exceeds the weakest valuation-8 frequency checked. The natural repair is to restrict to primitive frequencies, excising the whole non-primitive spectrum (every valuation

$c < l$, the pair above included) and controlling the remainder by an averaged bound; whether that remainder stays sparse enough at the scale that matters is a question this section answers computationally, for the family $R_{j-1,j}$ at $l = 18$, $j = 16$ (the first-order cardinality threshold, $T := \binom{32}{16}$, six budgets below the actual covering threshold $j^*(18) = 22$: coverage necessarily fails here already, and this level sits below the $j \geq l$ range Sections 6–7 otherwise work in), with a negative result. Write $\|S\|_1 := \sum_{\gcd(t,3)=1} |S(t)|/T$ for the normalized L^1 mass over frequencies coprime to 3. This restriction excludes $t = 3^{l-1}$, whose magnitude $|S(3^{l-1})| \sim T/\sqrt{3}$ is fixed by Proposition 9 and, empirically, is the largest at the specific levels this section checks directly (not in general, as the counterexample above shows), along with its conjugate $t = 2 \cdot 3^{l-1}$ (since $S(3^l - t) = \overline{S(t)}$, that frequency matches it exactly and is excluded for the same reason). The restriction to $\gcd(t, 3) = 1$ also excludes every nonprimitive frequency, exactly a third of the spectrum, entirely, so $\|S\|_1$ below measures only the primitive-frequency mass. Here $\|S\|_1 = 5226.01$. A fixed threshold makes the visible spikes look sparse, but of the 8,014 primitive frequencies with $|S(t)|/T$ exceeding 0.001, their combined contribution to $\|S\|_1$ is about 12.2, so over 99.7% of the primitive-frequency mass at this single computed level lives in an exponentially numerous population of coefficients individually below the threshold, with no claim here about how that mass is distributed within it. This is evidence against a sparse-exceptional-set repair among primitive frequencies at this accessible level; it says nothing about nonprimitive frequencies, about every l, and, since $j = 16$ sits six budgets below $j^*(18)$, still less about the mass distribution near the actual covering threshold.

The same $2^a 3^b$-type structure connects the problem to a 2011 discussion of Terence Tao's, applying Littlewood–Offord and Riesz-product methods to an object built from the same $2^a 3^b$-type exponential sums, in the closely related setting of a hypothetical non-trivial Collatz cycle [8]; his construction is structurally close to a boundary slice of $R_{j,k}$ with the bottom exponent pinned at 0. His own conclusion, in his own words, is that "once one uses the transcendence theory bound, ... Littlewood-Offord techniques are no longer of much use," and that he knows "of no way to get a nontrivial separation property between powers of 2 and powers of 3 other than via transcendence theory methods."

5.3. **A direct test of the residues that actually fail.** The most direct test examines the residues that actually fail to be covered, one budget step before covering succeeds, i.e. $H(l, j^*(l)-1)$ (Section 6). For $l = 10, \ldots, 15$ this set has size $2, 1, 3, 1, 3, 1$ respectively, tiny against a global occupancy mean $\lambda = |R_{j^*(l)-2,j^*(l)-1}|/(2 \cdot 3^{l-1})$ that is itself in the thousands over the same levels (from 1019 at $l = 10$ to 3695 at $l = 15$). Write the *depth-c cell* of a residue z for its class modulo 3^c, and its *depth-c local intensity* $\lambda_c(z)$ for the total hit count within that class divided by 3^{l-c}, the class's size (defined for $c < l$; at $c = l$ the class is $\{z\}$ itself, and $\lambda_l(z) = 0$ identically for any holdout, a degenerate case excluded below). A *non-homogeneous Poisson model* fit to that intensity assigns z hole probability $e^{-\lambda_c(z)}$. Computed directly from the exact histogram for $l = 10, \ldots, 15$, every one of the eleven holdouts across these six levels has λ_c well below the global mean at every depth $c = 8, 9, 10$ with $c < l$ (at $l = 10$, only $c = 8, 9$; λ_c from 2.0 to 108.4, hole probability from e^{-2} to e^{-108}). The non-homogeneous Poisson model assigns probability e^{-108} to an event that occurred; z, the levels, and the depths checked are all selected after seeing the holdouts, so this is not a predeclared, calibrated test: the exponent is suggestive rather than a formal rejection, and the model is not treated as a live candidate below on that basis. What survives is the weaker, descriptive claim underneath it: holdouts sit systematically in low-intensity cells, with how far below the mean varying widely by depth and level. At the finest depth the definition allows short of the degenerate case, $c = l - 1$ (a class of only 3 siblings), the same model uses the leave-one-out intensity $\lambda_{l-1}^{-z}(z) := \big(\text{sum of the other 2 siblings' hit counts}\big)/2$ in place of $\lambda_{l-1}(z)$, excluding z's own count and dividing by the remaining class size (not 3), to avoid the self-referential bias a class that small would otherwise carry, for every residue rather than only holdouts. Two separate checks follow from this, addressing two separate questions. First, does the model still look grossly refuted at this depth, the way the coarser depths above looked (hole probability down to e^{-108})? No: the expected total hole count under the model, summed over every unit, is 2.1 against 2 holdouts actually observed at $l = 10$, 3.5 against 1 at $l = 11$, 4.5 against 3 at $l = 12$, 5.3 against 1 at $l = 13$, 5.0 against 3 at $l = 14$ ($l = 15$ not run at this depth); this aggregate total is invariant to which residues the holes actually are, so it does not by itself say anything about whether intensity picks out the correct residues, only that the model is no longer grossly miscalibrated in aggregate at this depth (the two levels where the model most overpredicts, $l = 11$ and $l = 13$, are themselves evidence the fit is loose rather than tight). Second, does intensity pick out which specific residues resist? Here the answer is more informative: ranking every unit by this same leave-one-out intensity (ties, where the integer-valued sum coincides, broken arbitrarily but consistently, and immaterial to the conclusion since no holdout sits inside a large tied block), the actual holdouts sit deep in the low-intensity tail at every level checked this way (ranks 12 to 582 among 354,294 to 3,188,646 units, $l = 12, 13, 14$ only), though not always among the very lowest few residues by rank. It is this rank check, rather than the aggregate total above, that supports the correlation claim: read at this depth, local intensity correlates strongly with which residues resist, more strongly than the coarser depths $c = 8, 9, 10$ alone suggest, but does not fully determine it. A separate diagnostic at a smaller, more tractable pair, $(l, m) = (14, 16)$, for the family $R_{m-1,m}$ (mean hit count 188.5, three budgets below the covering threshold $j^*(14) = 19$), holds every $|S(t)|$ fixed and randomizes the phases of conjugate-symmetric pairs. Writing $N_l(z)$ for the exact hit count of residue z, N_{l-1} for the same tuple family's count one level down, and

$$F(z) := 3N_l(z) - N_{l-1}(z \bmod 3^{l-1})$$

for their imbalance among the three lifts of each parent residue, restricting to primitive frequencies $3 \nmid t$ gives exactly $\sum_{3\nmid t} S(t)e(-tz/3^l) = 3^{l-1}F(z)$, so F is identically 0 at every non-unit z (both $N_l(z)$ and $N_{l-1}(z \bmod 3^{l-1})$ vanish there, since no element of a tuple family is divisible by 3). Over all z, the actual ratio of this quantity's maximum absolute value to its root-mean-square is 15.79, while eight independent phase scrambles (fixed seed) give ratios in $[5.11, 5.24]$ (a second, independently seeded batch of eight gives the wider $[4.98, 5.40]$, so eight trials do not pin the null down tightly). The scrambles hold every $|S(t)|$ fixed but do not preserve the identically-zero-off-units constraint just derived, or integrality, so they are not the right null on their own.

The zero-off-units constraint has a direct algebraic meaning for the phases. Grouping primitive frequencies into triples $t = r,\ r + 3^{l-1},\ r + 2 \cdot 3^{l-1}$ by parent residue r modulo 3^{l-1} ($3 \nmid r$), the vanishing of F on the subgroup $3(\mathbb{Z}/3^l\mathbb{Z})$ is equivalent, by the same character-sum collapse used in Proposition 8, to $S(t_1) + S(t_2) + S(t_3) = 0$ for every such triple: three fixed magnitudes forming a closed triangle, non-degenerate for all but a negligible fraction (checked directly at $(l, m) = (14, 16)$: exactly two of the 1,062,882 primitive-parent triples, a conjugate pair, have the largest magnitude equal to the sum of the other two; the null's rotation-and-reflection construction below is only possibly less random on these two, with no visible effect on a 30-trial statistic). This pins the phases down to a one-parameter family, a common rotation of the actual configuration, together with its mirror image (negating all three phases before rotating, which also sums to zero). The conjugate partner of the triple at parent r is the triple at parent $3^{l-1} - r$, and $S(3^l - t) = \overline{S(t)}$ (used already in the proof of Proposition 8) links the two: only one triple per conjugate pair is given an independent random rotation and an independent coin flip for the reflection; the partner's phases are then fixed by conjugation rather than drawn on their own, so the resulting field stays real, matching F itself. This keeps every $|S(t)|$ fixed and vanishes off units exactly by construction, though not integrality (the actual array satisfies further constraints this null does not, e.g. $-N_{l-1}(r) \leq F(z) \leq 2N_{l-1}(r)$ per parent, and is nonetheless more extreme). Implemented and run at $(l, m) = (14, 16)$ (30 trials, checked into the repository, Section 9): the constrained null gives max /RMS ratios averaging 6.35, range $[5.93, 7.20]$, against the real array's 15.79, a residual factor of about 2.5. By Parseval, $\sum_z F(z)^2$ (hence the all-z mean square) is the same for the actual array and for any phase null preserving every $|S(t)|$, constrained or not, since it depends only on the magnitudes; a null respecting the zero-off-units property puts that same total squared mass on only the $2 \cdot 3^{l-1}$ unit positions instead of spreading it over all 3^l, which is exactly what forces $\mathrm{RMS}_{\mathrm{units}} = \sqrt{3/2} \cdot \mathrm{RMS}_{\mathrm{all}}$ for such a null. ($\mathrm{RMS}_{\mathrm{units}}$ itself enters no reported number below; every ratio quoted, actual, unconstrained, or constrained, uses $\mathrm{RMS}_{\mathrm{all}}$ throughout, identical across all three by Parseval, so the entire 6.35-vs-5.2 gap between the constrained and unconstrained nulls lives in the maximum, not in which RMS is used.) This identity governs only the RMS, not the maximum: Parseval says nothing about how concentrating the same squared mass onto fewer, mutually constrained positions affects an extreme-value statistic, and the triple correlations the construction introduces could in principle push max /RMS either direction. That the measured constrained-null ratio, 6.35, lands close to $\sqrt{3/2} \approx 1.22$ times the unconstrained scrambles' range is an empirical property of this specific construction, observed directly in the 30 trials rather than deduced from Parseval; the residual factor of 2.5 rests on these measured numbers rather than on the Parseval identity alone. Both the eight-trial unconstrained batches above and this thirty-trial constrained one are still too few to calibrate a rigorous significance level (a rank-based test against thirty draws cannot reach below $p \approx 1/31$, however extreme the observed value); what the residual factor near 2.5 provides, on its own, is evidence against the actual phases being fully exchangeable with a null that respects only $|S(t)|$ and the zero-off-units constraint, rather than a deductive demonstration (thirty draws too few to calibrate a significance level, as just noted, can suggest but not establish non-exchangeability). It does not, by itself, say that this departure needs phase structure specifically, since neither null above carries any information about the level-$(l-1)$ local intensity documented earlier in this section.

A third null tests the analogous parent-level information directly, for this array's own family: instead of fixing $|S(t)|$, fix each parent's raw total, $N_{l-1}(r)$ for every r, and split that total multinomially $(\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$ among its three lifts, carrying no information about the observed primitive-frequency phases (the split does still fix each parent's three lifts to sum to its total and is not a claim about the finer leave-one-out intensity used in the rank check above, a property of individual children rather than of the parent total this null actually conditions on). Run 30 times (checked into the repository, Section 9): this parent-total-conditioned null gives max /RMS ratios averaging 17.4, range $[14.9, 21.1]$, exceeding the actual array's 15.79 in about 83% of trials. The ratio alone would suggest parent-total randomness reproduces the observed extremity; it does not survive a look at the two absolute quantities the ratio divides. For each lift i of a parent total N under this null, n_i is Binomial$(N, \frac{1}{3})$, so the imbalance $3n_i - N$ has variance $2N$; summing the three lifts of every parent over the whole array gives $\mathbf{E}[\sum_z F(z)^2] = 6T$ exactly, $T = \binom{2m}{m}$. The actual array's $\sum_z F(z)^2$ is 9.35×10^9 against an expectation of $6T = 3.61 \times 10^9$, a factor of 2.59 and, against the null's own trial-to-trial spread over the 30 runs (standard deviation 5.6×10^6), about 1,030 null standard deviations out: this is the genuine departure, a magnitude excess, $|S(t)|$ itself departing from what independent splitting of the exact parent counts produces. The null's absolute maximum, averaging 477 and never exceeding 580 over the 30 trials, falls short of the actual array's 698 in every trial, but this adds no further evidence beyond the same excess: at the actual array's own root-mean-square (44.2 against the null's 27.5, a factor of 1.61), the null's mean maximum would scale to about $477 \times 1.61 \approx 768$, above the actual 698, so the maximum comparison is the root-mean-square shortfall restated, not an independent one. The max /RMS ratios themselves agree, or rather the null exceeds the actual ratio in most trials, precisely because both the maximum and the root-mean-square fall short by comparable factors (1.46 and 1.61); on this scale-free statistic the null is not refuted at all, and the extremity claim above rests entirely on the constrained phase null, at identical root-mean-square. (This uses the $R_{15,16}$ array's own parent totals directly, rather than the λ_c figures reported earlier in this section, which come from a different family and depth,

$R_{j^*(l)-2,j^*(l)-1}$ at $c = 8, 9, 10$; the two are separate measurements of the same qualitative phenomenon, coarse skew in tuple counts, rather than the same measurement.) So the multinomial null does not remove the basis for attributing the observed excess energy to structure beyond the parent totals; it is that excess. It says nothing about phase, which is the constrained null's own territory above: that null holds every $|S(t)|$ fixed by construction (Parseval then guarantees identical RMS for the actual array and the null alike) and still finds the actual maximum, 698, well above the constrained null's mean of about 280 in the same units (6.35 times the shared RMS of 44.2), a purely phase-driven departure the multinomial null cannot speak to since it never holds $|S(t)|$ fixed to begin with. Neither null, on its own or together, establishes which mechanism, if either, decides coverage, or how much of either departure a properly calibrated test would confirm. At the levels and pair checked, holdout rarity ($l = 10, \dots, 15$) and the phase-scramble gap ($(l, m) = (14, 16)$ alone) are not established as independent phenomena; both are plausibly connected to the same underlying skew in tuple counts.

Proposition 9. *For every $l \geq 1$, $|S(3^{l-1})|/\binom{2m}{m} \to 1/\sqrt{3}$ as $m \to \infty$, for the family $R_{m-1,m}$.*

Proof sketch. $\sum_i 2^{a_i} 3^i \equiv 2^{a_0} \pmod 3$, so $S(3^{l-1})/\binom{2m}{m}$ depends only on the limiting distribution of the top exponent a_0's parity as $m \to \infty$, which converges to the mixture $\frac{1}{3}\omega + \frac{2}{3}\omega^2$ of the two nontrivial cube roots of unity $\omega = e^{2\pi i/3}$ (the limiting gap $2m - 1 - a_0$ of the maximum of a random m-subset of $\{0, \dots, 2m-1\}$ is geometric with parameter $1/2$, giving $\Pr[a_0 \text{ even}] \to 1/3$). This mixture equals $-\frac{1}{2} - \frac{i\sqrt{3}}{6}$, of modulus $1/\sqrt{3}$. □

$t/3^l = 1/3$ exactly, so l drops out of $S(3^{l-1})$ entirely (only $V \bmod 3$ matters), and the limit above holds at every scale m, including along the covering-relevant diagonal $m \sim l \log_4(3)$. It sharpens Proposition 8's exact bound $|S(3^{l-1})| \geq T/2$ to the exact asymptotic ratio $1/\sqrt{3} \approx 0.577$, and confirms that the mass Proposition 8 forces concentrates, at this one frequency at least, exactly where the numerics of §5.2 place it: at the highest-valuation frequency, $t/3^l = 1/3$.

6. A theorem on the residues that resist coverage

Write $H(l, j)$ for the set of units modulo 3^l *not* covered by $R_{j-1,j}$. By definition of $j^*(l)$ as the least covering budget, $H(l, j^*(l)) = \emptyset$ and $H(l, j) \neq \emptyset$ for every $j < j^*(l)$; whether this extends to "$j \geq j^*(l)$ exactly when $H(l, j) = \emptyset$" needs emptiness to persist at every larger budget too, which is not automatic from the definition alone. Every result in this section needs $j \geq l$; $j^*(l) \geq l$ holds outright for $l = 1, \dots, 23$ ($l = 1$ from Table 1 directly, $l = 2, \dots, 23$ by Lemma 1 above; equality only at $l = 1$), but the conjectured rate $\log_4 3 < 1$ would eventually put $j^*(l)$ below l, outside this section's reach. Within this range, persistence does hold: Theorem 10 below gives $H(l, j) = \emptyset \Rightarrow H(l, j+1) \subseteq 2\emptyset \cap 4\emptyset = \emptyset$, so "$H(l, j) = \emptyset$ exactly when $j \geq j^*(l)$" is established for every $j \geq l$. We make no claim about $j < l$.

Theorem 10 (Holdout doubling inclusion)**.** *For $j \geq l$, $H(l, j+1) \subseteq 2H(l, j) \cap 4H(l, j)$.*

Proof. $R_{j-1,j}$ is the set of values $r = \sum_{i=0}^{j-1} 2^{b_i} 3^i$ over j-subsets $\{b_0 > \cdots > b_{j-1}\} \subseteq \{0, \dots, 2j-1\}$, and $R_{j,j+1}$ the same with $(j+1)$-subsets of $\{0, \dots, 2j+1\}$. Fix $r \in R_{j-1,j}$ with exponent set S, and $s \in \{1, 2\}$. Shifting every exponent up by s gives $S + s \subseteq \{s, \dots, 2j-1+s\} \subseteq \{1, \dots, 2j+1\}$, a j-subset whose induced value is $2^s r$ (a uniform shift of exponents preserves relative rank, so each term's power of 3 is unchanged). Since $S + s$ contains no 0, adjoining 0 as the new smallest exponent gives a $(j+1)$-subset of $\{0, \dots, 2j+1\}$: it becomes the new rank-j term, contributing $2^0 \cdot 3^j = 3^j$. So this witness represents $2^s r + 3^j$, and $3^j \equiv 0 \pmod{3^l}$ once $j \geq l$. Hence $2R_{j-1,j} \cup 4R_{j-1,j} \subseteq R_{j,j+1} \pmod{3^l}$ for $j \geq l$. So if $z \in H(l, j+1)$, i.e. z is not represented at budget $j+1$, then z lies in neither $2R_{j-1,j}$ nor $4R_{j-1,j}$: both $z/2$ and $z/4$ are unrepresented at budget j, i.e. $z \in 2H(l, j) \cap 4H(l, j)$. □

Corollary 11 (Chain contraction). *For $j \geq l$: if $x, 2x, \ldots, 2^{t-1}x \in H(l, j+1)$ are pairwise distinct (all reduced modulo 3^l, $t \geq 1$), then $2^i x \in H(l,j)$ for every $i = -2, \ldots, t-2$ (i.e. $x/4, x/2$ and, when $t \geq 2$, $x, 2x, \ldots, 2^{t-2}x$), pairwise distinct, a chain of length $t+1$.*

Proof. Apply Theorem 10 to each of the t elements: $2^i x \in H(l, j+1)$ gives $2^{i-1}x, 2^{i-2}x \in H(l,j)$ for $i = 0, \ldots, t-1$. The union of these pairs over all i is exactly $\{2^i x : i = -2, \ldots, t-2\}$, $t+1$ elements indexed by $t+1$ consecutive powers of 2. Since 2 has order $2 \cdot 3^{l-1}$ modulo 3^l (an elementary lifting-the-exponent fact, used already in Section 4), any $2 \cdot 3^{l-1}$ consecutive powers of 2 times x exhaust every unit modulo 3^l, and any shorter run is pairwise distinct. The hypothesis that $x, 2x, \ldots, 2^{t-1}x$ are pairwise distinct already forces $t \leq 2 \cdot 3^{l-1}$, so it remains to rule out equality. If $t = 2 \cdot 3^{l-1}$, those t elements would be every unit modulo 3^l, including every element of the (nonempty) image of $R_{j,j+1}$; but $H(l, j+1)$, the complement of that image, contains no such element. Hence $t < 2 \cdot 3^{l-1}$, and the $t+1$ output elements, a shorter run of consecutive powers, are pairwise distinct. ☐

Corollary 12 (Run-length bootstrap). *Writing* $\operatorname{maxrun}(H)$ *for the greatest t such that H contains t pairwise distinct elements of the form $x, 2x, \ldots, 2^{t-1}x$ (all reduced modulo 3^l), with* $\operatorname{maxrun}(\emptyset) := 0$, *$j^*(l) \leq j + \operatorname{maxrun}(H(l,j))$ for every $j \geq l$.*

Proof. Let $r := \operatorname{maxrun}(H(l,j))$ and suppose, for contradiction, $H(l, j+r) \neq \emptyset$; pick $z \in H(l, j+r)$, a chain of length 1 ($t = 1$) in the sense of Corollary 11. Applying Corollary 11 once at each of the r budgets $j+r-1, j+r-2, \ldots, j$ (all $\geq l$, since $j \geq l$), a chain of length t at budget $j'+1$ becomes a chain of length $t+1$ at budget j' each time, so after r applications the length-1 chain at $j+r$ becomes a chain of length $r+1$ in $H(l,j)$, contradicting $\operatorname{maxrun}(H(l,j)) = r$. So $H(l, j+r) = \emptyset$, i.e. $j^*(l) \leq j + r = j + \operatorname{maxrun}(H(l,j))$. ☐

Empirical Result 13. Corollary 12 is empirically exact, $j^*(l) = j + \operatorname{maxrun}(H(l,j))$, at every (l,j) with $l+1 \leq j \leq j^*(l)$ we were able to compute: $l = 2$ checked directly at its only nontrivial budget, $j = l+1 = 3$ ($H(2,3) = \{7\}$ modulo 9, maxrun $= 1$, matching $j^*(2) = 4$), $l = 3, 4$ checked directly the same way at $j = l+1$ ($\operatorname{maxrun}(H(3,4)) = 2$ and $\operatorname{maxrun}(H(4,5)) = 2$, matching $j^*(3) = 6$ and $j^*(4) = 7$, with the boundary-width argument following Proposition 24 below also proving the same equality there as a theorem rather than an empirical check), $l = 5, \ldots, 21$ at every such budget, and $l = 22$ at the single budget $j = l+1$, including a case with $|H(l,j)| > 3 \times 10^6$. (Past $j^*(l)$, $H(l,j) = \emptyset$ and $\operatorname{maxrun}(\emptyset) = 0$, so the displayed equality no longer holds there; the claim is only about budgets up to and including $j^*(l)$.) The lower endpoint $j = l+1$ has a real justification behind it: equality at $j = l$ itself holds for $l = 1, \ldots, 5$ but fails at every level checked, $l = 6, \ldots, 9$ ($\operatorname{maxrun}(H(6,6)) = 5$, giving $j + \operatorname{maxrun} = 11 \neq 10 = j^*(6)$; $\operatorname{maxrun}(H(7,7)) = 5$, giving $12 \neq 11 = j^*(7)$; $\operatorname{maxrun}(H(8,8)) = 8$, giving $16 \neq 12 = j^*(8)$; $\operatorname{maxrun}(H(9,9)) = 8$, giving $17 \neq 13 = j^*(9)$); Corollary 11 and Corollary 12 both need $j \geq l$ to apply at all, but that alone does not explain why $j = l$ specifically starts failing at $l = 6$. The sub-range $l+2 \leq j \leq j^*(l)$ is proven at $l = 3, \ldots, 13$: Proposition 24 below, which also proves the boundary budget $j = l+1$ at $l = 3, 4, 5, 6$ specifically. From $l = 7$ on, $j = l+1$ remains empirical only, tied to Proposition 22's own open converse.

A stronger, per-witness converse of Theorem 10, that $x/2, x/4 \in H(l,j)$ already forces $x \in H(l, j+1)$, which would in turn make a maximal doubling chain both sufficient and necessary for survival to a given budget, is false as an unrestricted statement: $1547 \in 2H(7,8) \cap 4H(7,8)$, yet $1547 \notin H(7,9)$ (both of 1547's two budget-9 witnesses have an exponent configuration that Theorem 10's shift-and-adjoin construction does not directly reconstruct from a budget-8 witness). Section 7 returns to a restricted, weaker form of this converse (corner-redundancy, not the per-witness statement this counterexample refutes) that remains open in general and is proven at specific levels.

A natural strengthening of Theorem 10, that every individual witness admits a bounded-cost local repair, is also false in its most optimistic form.

Call the *Durfee depth* of a witness $\{a_0 > \cdots > a_{j-1}\} \in R_{j-1,j}$ at budget j the count $\#\{i : a_i \geq j\}$, the number of chosen exponents at or above the budget itself.

Empirical Result 14 (Failure of the cost-1 local repair rule). At the budget transition $(l, j) = (6, 10) \to (7, 11)$, call a unit modulo 3^7 a *child* and its reduction modulo 3^6 its *parent*; 100 of the 1458 children (all units modulo 3^7) have minimum achievable Durfee depth exceeding their parent's minimum depth by more than one, up to three; at $(7, 11) \to (8, 12)$, 332 of 4374 children (all units modulo 3^8) do, likewise up to three. A correspondingly exhaustive check defines the *repair cost* of a child residue r with parent residue p as $\min_{S,S'} |S \triangle S'|$, the smallest symmetric difference of exponent sets over every witness exponent set S for p at budget j and every witness exponent set S' for r at budget $j + 1$ ($|S| = j$, $|S'| = j + 1$, so this distance is always odd, ≥ 1, and the minimum ranges over the full witness fiber at each budget). This finds repair costs up to 7 at both transitions, against a cost of 1 for the overwhelming majority. Since every child at both transitions is checked exhaustively, this falsifies a repair cost of exactly 1 at these two transitions outright; it does not establish that no bounded repair cost exists.

This falsifies, in its originally proposed cost-1 form, a natural conjecture this project had earlier entertained (that coverage propagates via a uniform, cost-1 local repair rule budget by budget); its valid surviving content is Theorem 10 and its corollaries above.

7. Further structure of the last holdouts

The residues in $H(l, j^*(l) - 1)$, the last to be covered, satisfy several further exact laws. For $W \geq l - 1$, write

$$U(l, W) := \left\{ \textstyle\sum_{i=0}^{l-1} 2^{\beta_i} 3^i \bmod 3^l \;\middle|\; W \geq \beta_0 > \cdots > \beta_{l-1} \geq 0 \right\},$$

an l-term analogue of $R_{j-1,j}$ built from a fixed term count and a growing exponent ceiling W.

Lemma 15 (Width scaling identity). *For $j \geq l \geq 1$,*

$$R_{j-1,j} \equiv 2^{j-l}\, U(l, j + l - 1) \pmod{3^l}. \tag{4}$$

Proof. Since $3^i \equiv 0 \pmod{3^l}$ for $i \geq l$, the residue of $r = \sum_{i=0}^{j-1} 2^{b_i} 3^i \in R_{j-1,j}$ modulo 3^l depends only on its top l exponents $b_0 > \cdots > b_{l-1}$. A decreasing l-tuple $b_0 > \cdots > b_{l-1} \geq 0$ with $b_0 \leq 2j - 1$ extends to a full admissible j-tuple in $\{0, \ldots, 2j - 1\}$ (appending $j - l$ further, strictly smaller, distinct exponents) exactly when $b_{l-1} \geq j - l$: append $b_{l-1} - 1, \ldots, b_{l-1} - (j - l)$ when this holds, and no extension exists when it does not, since fewer than $j - l$ nonnegative integers lie below b_{l-1}. Writing $c_i := b_i - (j - l)$ turns $b_{l-1} \geq j - l$ into $c_{l-1} \geq 0$ and $b_0 \leq 2j - 1$ into $c_0 \leq 2j - 1 - (j - l) = j + l - 1$, and $\sum_i 2^{b_i} 3^i = 2^{j-l} \sum_i 2^{c_i} 3^i$. So the top-$l$-exponent tuples realizable by $R_{j-1,j}$ are exactly 2^{j-l} times the decreasing l-tuples counted by $U(l, j + l - 1)$, giving the claimed identity modulo 3^l. $\square$

Proposition 16 (Existence). *$j^*(l)$ exists for every $l \geq 1$, with $j^*(l) \leq 2 \cdot 3^{l-1} - 1$.*

Proof. $U(l, \cdot)$ is monotone increasing in W: enlarging the range $\{0, \ldots, W\}$ only adds admissible exponent tuples. Shifting every exponent of a tuple realizing $u \in U(l, W)$ up by 1 gives a decreasing l-tuple with top exponent at most $W + 1$ and bottom exponent at least $1 > 0$, still admissible, realizing $2u$; hence $2\,U(l, W) \subseteq U(l, W + 1)$ for every $W \geq l - 1$. The tuple $\beta_i = l - 1 - i$ ($i = 0, \ldots, l - 1$) is admissible at $W = l - 1$ and gives $u_0 := \sum_i 2^{l-1-i} 3^i \equiv 2^{l-1} \pmod 3$, a unit, so $u_0 \in U(l, l - 1)$. Iterating the inclusion, $2^k u_0 \in U(l, l - 1 + k)$ for every $k \geq 0$. Since 2 has order $2 \cdot 3^{l-1}$ modulo 3^l (already used in Section 4), $\{2^k u_0 : 0 \leq k < 2 \cdot 3^{l-1}\}$ is the coset $u_0\langle 2 \rangle = (\mathbb{Z}/3^l\mathbb{Z})^*$: every unit modulo 3^l appears as $2^k u_0$ for some k in that range, hence lies in $U(l, l - 1 + k) \subseteq U(l, W^\dagger)$, $W^\dagger := l + 2 \cdot 3^{l-1} - 2$, by monotonicity. So $U(l, W^\dagger) = (\mathbb{Z}/3^l\mathbb{Z})^*$, and Lemma 15 at $j = W^\dagger - l + 1 = 2 \cdot 3^{l-1} - 1 \geq l$ gives $R_{j-1,j} \equiv (\mathbb{Z}/3^l\mathbb{Z})^* \pmod{3^l}$, hence $j^*(l) \leq 2 \cdot 3^{l-1} - 1$. $\square$

This bound is exponential, far above the conjectured $\log_4 3 \cdot l$ rate; it establishes only that $j^*(l)$ is a well-defined finite quantity, which the rest of this section and Section 4's mean-payoff-game argument then bound far more tightly.

By Lemma 15, which itself needs $j \geq l$, the smallest $j \geq l$ with $U(l, j+l-1) = (\mathbb{Z}/3^l\mathbb{Z})^*$ is the smallest $j \geq l$ at which $R_{j-1,j}$ covers; since $j^*(l) \geq l$ for $l = 1, \dots, 23$ ($l = 1$ from Table 1 directly, $l = 2, \dots, 23$ by Lemma 1), with equality only at $l = 1$, this coincides with $j^*(l)$ itself throughout the range this paper works in, though nothing here rules out coverage at some unchecked $j < l$ for a level not yet computed. Writing $W^*(l) := j^*(l) + l - 2$ for the width one step before that,

$$H(l, j) = 2^{j-l}\, D(l, j+l-1), \qquad D(l, W) := (\mathbb{Z}/3^l\mathbb{Z})^* \setminus U(l, W). \tag{5}$$

Theorem 17 (Last-holdout parity). *If $j^*(l) \geq l+1$, every element of $H(l, j^*(l)-1)$ is $\equiv 1 \pmod 3$. This covers $l = 2, \dots, 23$ (Table 1); the hypothesis is inapplicable only at $l = 1$ ($j^*(1) = 1$, outside (5)'s range), which says nothing about whether the conclusion itself would hold there, and would become inapplicable again for any l at which the conjectured rate $\log_4 3 < 1$ eventually pushes $j^*(l)$ below $l + 1$.*

Proof. $U(l, \cdot)$ is monotone increasing in W, so $D(l, \cdot)$ is monotone decreasing; every $x \in D(l, W^*(l))$ is, by minimality of $j^*(l)$, representable at width $W^*(l) + 1$ but not at $W^*(l)$. Any witness for x at width $W^*(l) + 1$ using only exponents $\leq W^*(l)$ would already witness x at that smaller width, so every such witness has top exponent $\beta_0 = W^*(l) + 1$. Its contribution to the value, at rank 0, is $2^{\beta_0} \cdot 3^0 = 2^{W^*(l)+1}$, so $x \equiv 2^{W^*(l)+1} \equiv (-1)^{W^*(l)+1} \pmod 3$. By (5), which needs $j^*(l) - 1 \geq l$, elements of $H(l, j^*(l)-1)$ are $2^{j^*(l)-1-l}x$ for such x, hence

$$2^{j^*(l)-1-l}x \equiv (-1)^{j^*(l)-1-l} \cdot (-1)^{W^*(l)+1} = (-1)^{2j^*(l)-2} = 1 \pmod 3,$$

using $W^*(l) + 1 = j^*(l) + l - 1$ and that $2j^*(l) - 2$ is always even. ☐

Proposition 18 (Near-extinction: forward containment). *If $j^*(l) \geq l+2$, $H(l, j^*(l)-1) \subseteq 2 \cdot \{x \in H(l, j^*(l)-2) : x \equiv 2 \,(\mathrm{mod}\ 3)\}$.*

Proof. Write $J := j^*(l)$. Theorem 10 at $j' = J - 2 \geq l$ gives $H(l, J-1) \subseteq 2H(l, J-2)$, so every $y \in H(l, J-1)$ equals $2x$ for a (unique, since 2 is a unit) $x \in H(l, J-2)$. By Theorem 17, $y \equiv 1 \pmod 3$; since $y = 2x$, $2x \equiv 1 \pmod 3$, i.e. $x \equiv 2 \pmod 3$. So x lies in $\{x \in H(l, J-2) : x \equiv 2 \pmod 3\}$ and $y = 2x$ lies in twice that set. ☐

Empirical Result 19 (Near-extinction bijection). $H(l, j^*(l)-1) = 2 \cdot \{x \in H(l, j^*(l)-2) : x \equiv 2 \pmod 3\}$, exactly, verified at every $l = 2, \dots, 9$ against the exact table with no exception. Proposition 18 proves the $\subseteq$ direction unconditionally (at every level with $j^*(l) \geq l+2$, true throughout this range); only the reverse containment, that every $x \equiv 2 \pmod 3$ in $H(l, j^*(l)-2)$ actually survives via $2x$, remains empirical.

The same width mechanism that proves Theorem 17 governs the reverse containment one step earlier, through the second-highest exponent β_1 of the extremal witness; unlike the top-exponent argument above, pinning β_1 down needs the one-step width identity of Lemma 23 below. We have not carried that argument through to a full proof of this direction and record it as empirically verified only.

A natural cross-level congruence relating $H(l+1, j^*(l+1)-1)$ to $H(l, j^*(l)-1)$ does not hold. Writing π_l for reduction modulo 3^l, the candidate congruence $x \equiv 4\,\pi_l(x) \pmod{3^{l+1}}$ is impossible whenever $\pi_l(x)$ is a unit and $l \geq 2$: reducing both sides modulo 3^l forces $\pi_l(x) \equiv 4\,\pi_l(x) \pmod{3^l}$, i.e. $3\,\pi_l(x) \equiv 0 \pmod{3^l}$. Checked exhaustively at $l = 2, \dots, 6$, $\pi_l(x)$ never lies in $H(l, j^*(l)-1)$ for $x \in H(l+1, j^*(l+1)-1)$ in the first place, so the pattern has no witnessing instance at these levels either. No cross-level relation between $H(l+1, \cdot)$ and $H(l, \cdot)$ is established by this paper.

Proposition 20 (Mod-9 two-class containment). *For $l \geq 2$, writing $J := j^*(l)$, if $J \geq l+1$ then $H(l, J-1) \bmod 9 \subseteq \{4^J, 4^{J+1}\} \pmod 9$.*

Proof. As in Theorem 17's proof, every witness of $x \in D(l, W^*(l))$ at width $W^*(l)+1$ has top exponent $\beta_0 = W^*(l)+1$; modulo 9, only the rank-0 and rank-1 terms survive (rank $i \geq 2$ contributes a multiple of 3^2), so $x \equiv 2^{\beta_0} + 3 \cdot 2^{\beta_1} \pmod 9$ for the witness's second-highest exponent β_1. By (5), elements of $H(l, J-1)$ are $y = 2^{J-1-l}x$, so $y \equiv 2^{J-1-l+\beta_0} + 3 \cdot 2^{J-1-l+\beta_1} \pmod 9$; using $\beta_0 = W^*(l)+1 = J+l-1$, the first term is $2^{2J-2} = 4^{J-1} \bmod 9$, fixed. The second term, $3 \cdot 2^m \pmod 9$ for $m := J-1-l+\beta_1$, depends only on the parity of m (since $2^m \bmod 3$ has period 2), taking exactly the two values 3 or 6 $\pmod 9$ regardless of β_1's actual value; a direct check gives $\{4^{J-1}+3, 4^{J-1}+6\} \equiv \{4^J, 4^{J+1}\} \pmod 9$ for every J. So $y \bmod 9 \in \{4^J, 4^{J+1}\}$. ☐

Empirical Result 21 (Exclusion of the lower class). The class actually occupied is 4^{J+1}, never 4^J, at every level checked, $l = 2, \ldots, 16$. Excluding 4^J in general is the sharpest concrete open question this part of the investigation produced: Proposition 20 places no constraint on β_1 itself, only on the parity-dependent outcome, so ruling out one of the two parities needs a different argument.

Proposition 22 (Sufficiency of bounded maxrun)**.** *If* $\operatorname{maxrun}(H(l, l+1))$ *is bounded by a constant* C *for all* l*, then* $j^*(l) \leq l + C + 1$*.*

Proof. Corollary 12 at $j = l+1$. ☐

The converse is not established, and the natural argument for it does not work: reading Corollary 12 downward gives only $\operatorname{maxrun}(H(l, l+1)) \geq j^*(l) - l - 1$, a *lower* bound on maxrun, which does not force maxrun to stay bounded when $j^*(l) - l$ does. Proving the converse would need an upper bound on maxrun in terms of $j^*(l) - l$, equivalent to a tightness statement for Corollary 12 that this section's own corner-redundancy question below addresses and that is false in the strong, per-witness form: recall 1547 from Section 6.

A naive expectation, that a random set of holdouts at the observed density would have doubling chains far longer than the flat ceiling of 3–4 measured at every level $l = 5, \ldots, 22$, does not survive contact with the actual, rapidly declining density of $H(l, l+1)$. An independence model matched to that actual density predicts values in the same narrow range the observed sequence occupies, without the flagrant mismatch a naive doubling-chain estimate would predict. That range is itself narrow, only 3 or 4 over the entire computed range $l = 5, \ldots, 22$, so this comparison mainly rules out gross overprediction by the naive model; it does not, on its own, establish that the independence model tracks the specific level at which the observed sequence rises from 3 to 4 and falls back, only that its predicted values stay inside that same narrow band. This comparison stays descriptive rather than a specified statistical test (no distributional family or wraparound convention is fixed), and does not, by itself, say anything about whether $\operatorname{maxrun}(H(l, l+1))$ is bounded; Proposition 22 is the only established link between that question and $j^*(l)$'s rate.

Lemma 23 (One-step width identity)**.** *Let* $\operatorname{Corner}(l, W+1)$ *be the set of values realized by a witness tuple* $\beta_0 > \cdots > \beta_{l-1}$ *with* $\beta_0 = W+1$ *and* $\beta_{l-1} = 0$ *simultaneously. Then*

$$U(l, W+1) = U(l, W) \cup 2U(l, W) \cup \operatorname{Corner}(l, W+1).$$

Proof. $\supseteq$: $U(l, W) \subseteq U(l, W+1)$ by monotonicity, $2U(l, W) \subseteq U(l, W+1)$ by the shift argument of Proposition 16's proof, and $\operatorname{Corner}(l, W+1) \subseteq U(l, W+1)$ since a corner witness is, in particular, a witness at width $W+1$.

$\subseteq$: let $u \in U(l, W+1)$, witnessed by $W+1 \geq \beta_0 > \cdots > \beta_{l-1} \geq 0$. If $\beta_0 \leq W$, the tuple already witnesses u at width W, so $u \in U(l, W)$. If $\beta_0 = W+1$ and $\beta_{l-1} \geq 1$, subtracting 1 from every exponent gives a decreasing tuple in $[0, W]$ (top exponent W, bottom exponent ≥ 0) witnessing $u/2$, so $u \in 2U(l, W)$. If $\beta_0 = W+1$ and $\beta_{l-1} = 0$, this is exactly the corner case, so $u \in \operatorname{Corner}(l, W+1)$. These three cases are exhaustive. ☐

At every width W in the range $2l+1 \leq W \leq j^*(l) + l - 1$, checked exhaustively for $l = 3, \ldots, 13$ (the upper end is the largest width the exact table reaches at each level), $\operatorname{Corner}(l, W+1)$ contributes nothing new: every corner witness's value is already covered without it. Call this the *corner-redundancy* property at width W; whether it holds at every width $W \geq 2l+1$ in general, beyond the levels checked, is open. It also holds, checked directly, at the one smaller width $W = 2l$ itself for $l = 3, 4, 5, 6$; from $l = 7$ through 13 it is checked to fail there. The same exhaustive check also covers every width strictly below $2l$, down to $W = l-1$, at every $l = 3, \ldots, 13$, though no result in this paper depends on that wider range. $W = 2l$ is exactly $j = l+1$, Proposition 22's own case; corner-redundancy at that width is known, one way or the other, at every level $l = 3, \ldots, 13$, holding only for $l = 3, 4, 5, 6$. What remains open is Proposition 22's own converse itself: corner-redundancy's failure at $W = 2l$ for $l \geq 7$ does not settle it either way, it only means this particular mechanism does not prove it there.

Proposition 24 (Corner-redundancy implies tightness)**.** *Fix j with $l+2 \leq j \leq j^*(l)$. If corner-redundancy holds at every width $2l+1 \leq W \leq j^*(l)+l-1$, then $j^*(l) = j + \operatorname{maxrun}(H(l,j))$ exactly.*

Proof. Corner-redundancy at width W says $\operatorname{Corner}(l, W+1) \subseteq U(l,W) \cup 2U(l,W)$, so Lemma 23 gives $U(l, W+1) = U(l,W) \cup 2U(l,W)$. Multiplication by 2 is a bijection of $(\mathbb{Z}/3^l\mathbb{Z})^*$, so it commutes with complementation and with intersection; taking complements turns this into $D(l, W+1) = D(l,W) \cap 2D(l,W)$, and pushing through the shift in (5) (itself a bijection, for the same reason) turns the width-level equality at $W = j' + l - 1$ into an equality of holdout sets, $H(l, j'+1) = 2H(l,j') \cap 4H(l,j')$, sharpening Theorem 10's inclusion to an identity whenever corner-redundancy holds at that width.

Claim: for every j' with $j \leq j' < j^*(l)$, $\operatorname{maxrun}(H(l,j'+1)) = \operatorname{maxrun}(H(l,j')) - 1$, given corner-redundancy at width $j'+l-1$. Since $j' < j^*(l)$, minimality of $j^*(l)$ gives $H(l,j') \neq \emptyset$, so $s := \operatorname{maxrun}(H(l,j')) \geq 1$ throughout. The inequality $\leq$ is unconditional: if $H(l,j'+1) = \emptyset$ it reads $0 \leq s-1$, true since $s \geq 1$; otherwise a maximal chain of length $\operatorname{maxrun}(H(l,j'+1)) \geq 1$ in $H(l,j'+1)$ extends, by Corollary 11, to a chain one longer in $H(l,j')$. For $\geq$, fix a maximal chain $x, 2x, \ldots, 2^{s-1}x \in H(l,j')$. For each $k = 2, \ldots, s$, both $2^{k-1}x$ and $2^{k-2}x$ lie in this chain, so the equality $H(l,j'+1) = 2H(l,j') \cap 4H(l,j')$ places $2^k x \in H(l,j'+1)$; this gives the $s-1$ consecutive powers $4x, 8x, \ldots, 2^s x \in H(l,j'+1)$, so $\operatorname{maxrun}(H(l,j'+1)) \geq s-1$, proving the claim.

The widths this uses, $j'+l-1$ for $j \leq j' < j^*(l)$, all lie in $[j+l-1, j^*(l)+l-2] \subseteq [2l+1, j^*(l)+l-1]$ (using $j \geq l+2$), inside the hypothesis, regardless of $\operatorname{maxrun}(H(l,j))$'s actual value. Telescoping the claim over every step from $j' = j$ to $j' = j^*(l)-1$ (a fixed number of steps, $j^*(l)-j$, fixed before any of this) gives $\operatorname{maxrun}(H(l,j^*(l))) = \operatorname{maxrun}(H(l,j)) - (j^*(l)-j)$. The left side is 0 ($H(l,j^*(l)) = \emptyset$ by definition), so $\operatorname{maxrun}(H(l,j)) = j^*(l) - j$, i.e. $j^*(l) = j + \operatorname{maxrun}(H(l,j))$. ☐

Since corner-redundancy is verified at every width the proof above uses for $l = 3, \ldots, 13$, Proposition 24 is unconditional at those levels: Empirical Result 13's equality is a proven theorem there, for every j with $l+2 \leq j \leq j^*(l)$. In general, whether this implication's hypothesis holds is exactly the corner-redundancy question above.

At $l = 3, 4, 5, 6$, where corner-redundancy also holds at the boundary width $W = 2l$, the same argument (starting the telescoping at $j' = l+1$ instead of $j' \geq l+2$) proves the boundary case too: $j^*(l) = (l+1) + \operatorname{maxrun}(H(l,l+1))$ exactly, verified directly ($l = 3$: $4+2 = 6$; $l = 4$: $5+2 = 7$; $l = 5$: $6+3 = 9$; $l = 6$: $7+3 = 10$). This is the per-level identity behind Proposition 22's own converse, proven at these four levels specifically; the converse itself is a claim about every l, and this route to it needs the identity at every level, these four included (an upper bound on $\operatorname{maxrun}(H(l,l+1))$ weaker than exact equality would also suffice, per the discussion above, but no such bound is established here). At $l = 7, \ldots, 13$, corner-redundancy at $W = 2l$ is checked to fail, so this route does not reach the converse there; beyond $l = 13$, corner-redundancy at $W = 2l$ is itself unchecked. The converse remains open at every $l \geq 7$.

8. Discussion

Two named questions summarize what stands between this paper's results and a sharper theorem. Excluding the lower class in Proposition 20's containment (Empirical Result 21) would turn the observed single-class pattern into a fully proven statement. Proving corner-redundancy at every width $W \geq 2l+1$ would, by Proposition 24, make Corollary 12's bootstrap tight for every budget $l+2 \leq j \leq j^*(l)$ in general, not merely at the levels $l = 3, \dots, 13$ where corner-redundancy is already verified; it would not, by itself, settle Proposition 22's converse at every level, which turns on the boundary budget $j = l+1$ ($W = 2l$), established above only at $l = 3, 4, 5, 6$, where corner-redundancy holds there too; at $l = 7, \dots, 13$, where it is checked to fail there, the converse remains open by this mechanism, and beyond $l = 13$ both the property and the converse are open. Neither question is, on its face, as hard as the Weak Covering Conjecture itself; whether either is much easier is not known.

A finite exponential sum built from strictly decreasing, exponentially weighted digit tuples is a natural object beyond this specific covering problem: finite-population L-statistics at non-CLT frequencies and weights, Littlewood–Offord theory for structured coefficient sequences, and the analytic side of the Collatz literature via Tao's discussion are all adjacent areas. Findings here may recur there under different names. The full-spectrum, uniform-cancellation route this paper tries is unreachable outright (Proposition 8); stronger, frequency-dependent cancellation strategies remain untried. Exceptional mass does not concentrate on a sparse set at the one accessible scale checked ($l = 18$, $j = 16$, one threshold, primitive frequencies only). Local intensity, read at a fine enough depth, correlates strongly with which residues resist without fully determining it, and a separate phase-randomization diagnostic departs from a null respecting the frequency magnitudes and the zero-off-units constraint they satisfy, at the one pair checked, $(l, m) = (14, 16)$. A further null carrying no primitive-frequency phase information matches that diagnostic's ratio statistic, but does so only by falling short of the actual array's absolute maximum and root-mean-square by comparable factors in every trial run; read in the absolute units the ratio divides, this second null is the one that departs from the data, by about a thousand of its own standard deviations on total energy, a magnitude excess the ratio statistic is silent on. Neither null, on its own or together, establishes what decides coverage, or calibrates either departure to a significance level. Tao's later, unrelated result that almost every Collatz orbit attains an almost bounded value [3] works with a first-passage random variable over a different probabilistic model of the same map and does not bear on covering directly, but is the clearest instance of a new kind of technique changing what counts as tractable in this area, the kind of shift a resolution of either open question above would need. Predecessor-density bounds on the $3n+1$ orbit structure itself, most developed in Krasikov and Lagarias's difference-inequality approach [4] (a lower bound of $x^{0.84}$ on the count of integers below x whose orbit reaches 1), target the same broad question as Wirsching's own reduction from covering to positive predecessor density, by a different mechanism: orbit-level difference inequalities rather than the covering structure of one finite family $R_{j-1,j}$. Nothing in that literature currently transfers to $K(l)$ or $j^*(l)$ directly, but the shared target makes a transfer more plausible than an unrelated result would, and whether one exists is itself an open question.

The Weak Covering Conjecture itself, and the exact asymptotic rate of $e(l)$, remain open.

9. Code and data availability

The repository `https://github.com/faculdade/weak-covering-conjecture` holds the code and data behind every computation reported here, one folder per section. It contains the covering search that produced Table 1, the model comparison behind Table 2, the mean-payoff-game solver, the exponential-sum computations of Section 5, and the holdout computations of Sections 6 and 7, together with the stored holdout sets at the final budgets of each level $l = 5, \dots, 21$; the $\mathrm{maxrun}(H(l, l+1))$ computation behind Proposition 22's discussion, which reaches $l = 22$, lives as a directly runnable tool in its own folder. The twelve certificates (σ_k, ρ_k, h_k) behind Table 3 are stored there with a verifier that re-checks each of them against the inequality in Theorem 5's proof, without re-running the solver. Each folder's README gives what it checks, the command, the output to expect, and what the run costs. Most checks take seconds; the three largest levels of Table 1 do not. Every claim above matches commit `b69d0d3` of that repository; later commits may extend it further but should not change what this specific commit already contains.

Acknowledgments. AI assistance was used for textual review and translation.

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